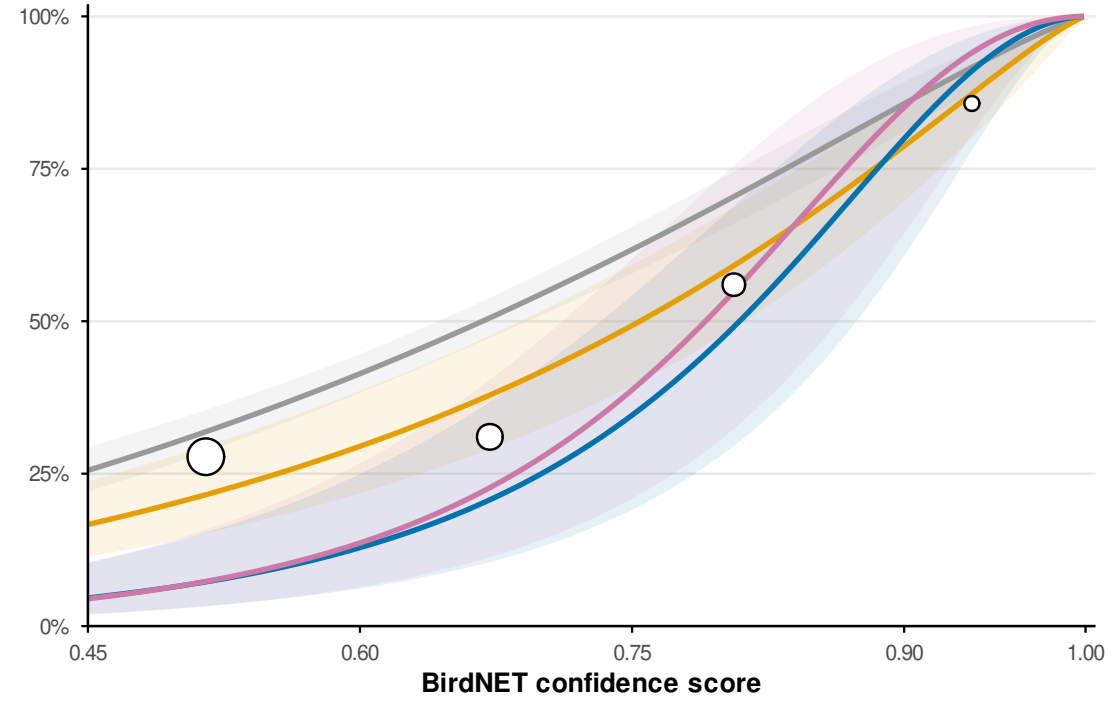
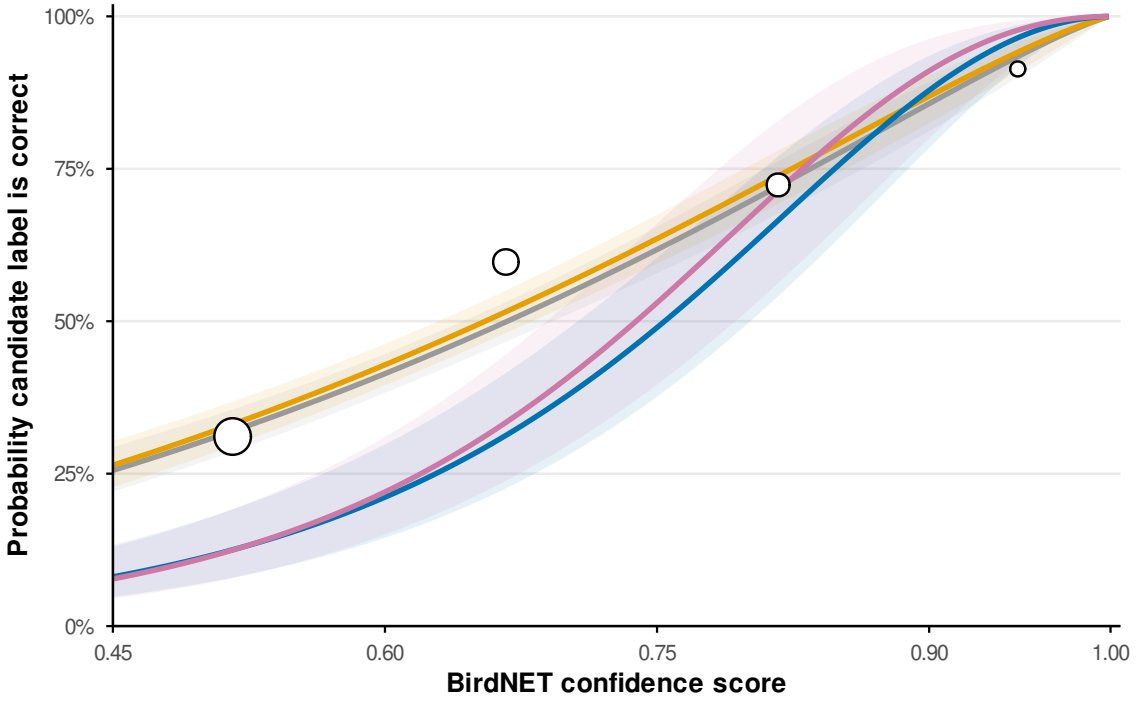


Probability candidate label is correct

A. Multi-candidate events



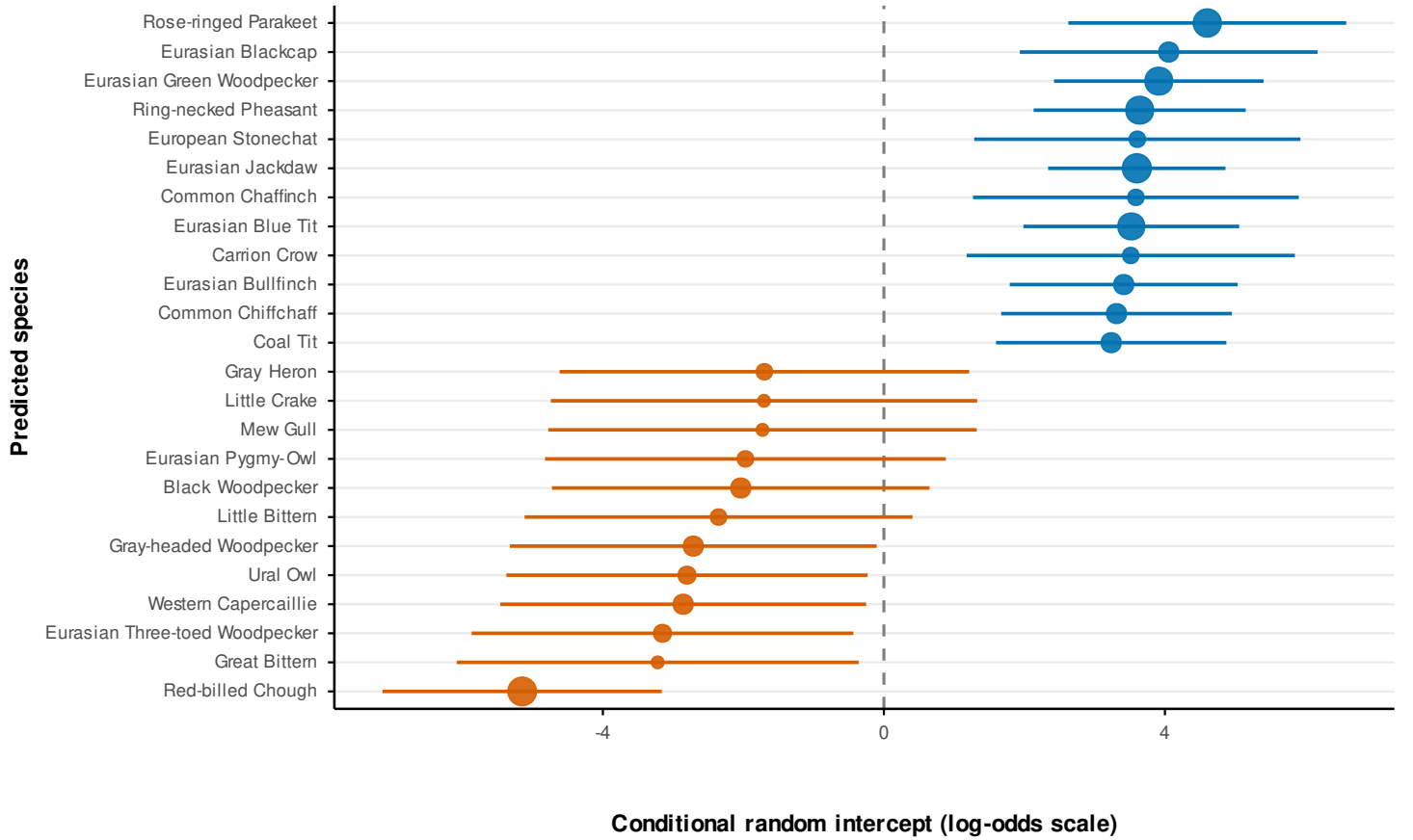
B. Single-candidate events



Model M1: Score M2: Score + event M3: Species-aware M4: Random score slope

Validated n 30 40 50 200 300 400 500

C. Largest Primary M3 species baseline deviations

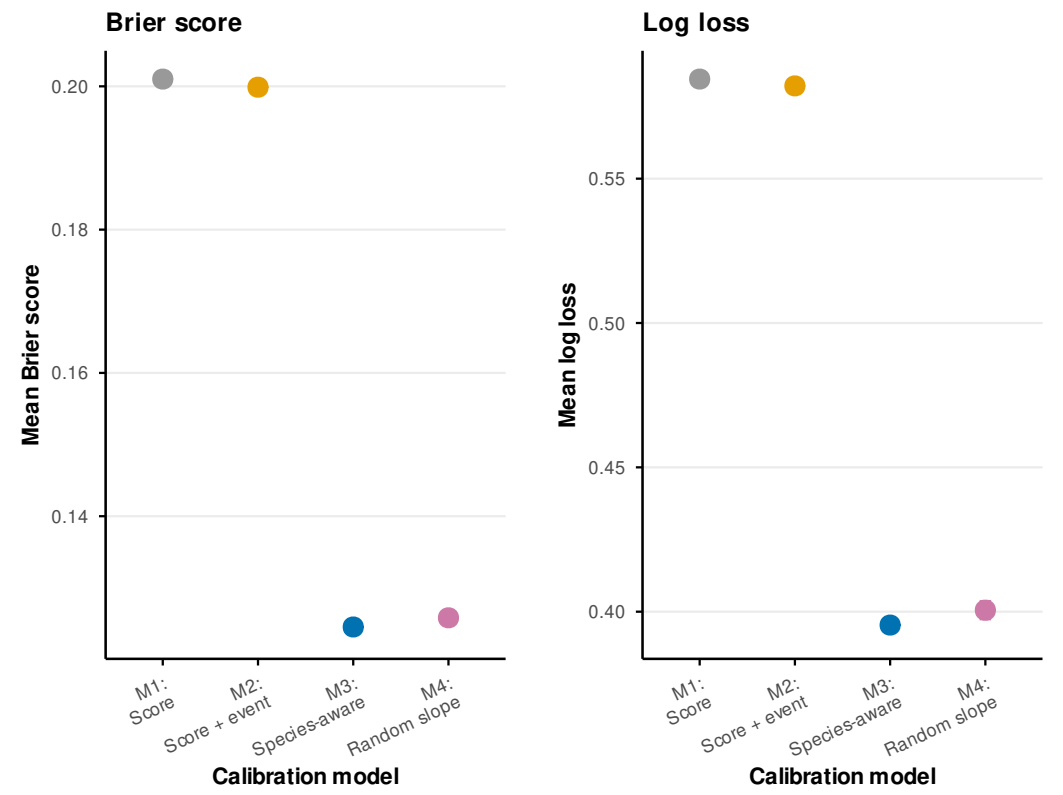


Predicted species

Conditional random intercept (log-odds scale)

Above population baseline Below population baseline Validated n 10 20 30 40

D. Event-grouped out-of-fold prediction error



Brier score

Log loss

Calibration model

Calibration model